

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 34)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Biomimetics uses knowledge of the adaptations of other species to improve the design of manufactured items.

0 1

1

Complete Table 1 by adding the missing adaptation or application.

[4 marks]

Table 1

Species and adaptation	Design application	
Tubercles (bumps) on the flippers of humpback whales		
	Self-cleaning glass	
Shark skin scales that reduce friction		
	Velcro	

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0 1

2

Explain how the design of termite mounds has been copied to reduce the environmental impact of large buildings.

[3 marks]

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0 1 . 3

Describe **two** ways in which studying the physiology of other species could help to solve human health problems.

[2 marks]

1

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2

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0 1 . 4

Explain why bird wing bones are a useful model for engineers.

[1 mark]

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10

0 2 . 1

For each of the following investigations, name the most appropriate statistical test.

[4 marks]

Whether noise level decreases as distance from a road increases

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Whether the numbers of dormice differ between woodlands with different management ..

.....

Whether mean dissolved oxygen differs between two streams

.....

Whether crop yield increases with fertiliser application (ranked data)

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0 2 . 2

Explain why the chi-squared test cannot be used for data such as wind speed or plant height.

[1 mark]

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0 2 . 3

A Spearman's rank test on 12 pairs of data gave $r_s = -0.72$. The critical value at $p = 0.05$ for $n = 12$ is 0.59. Explain what can be concluded.

[3 marks]

02 . 4

Explain the purpose of error bars on a graph of mean values.

[2 marks]

10

03 . 1

Define each of the following ecological terms and give an example.

[8 marks]

Species

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Ecological niche

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Community

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Biome

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03 . 2 Explain the difference between an ecosystem and a biome.

[2 marks]

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10

0 4

The Hawaiian goose (nene) was reduced by hunting, habitat loss and introduced predators to about 30 birds. A captive breeding programme started with just seven individuals and has increased the wild population to over 2000.

0 4

1

Explain why starting the programme with seven individuals caused problems, and describe how these problems are managed.

[4 marks]

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0 4

2

Explain **three** reasons, other than gene pool size, why species kept in captivity may fail to breed.

[3 marks]

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0 4

3

Describe how artificial insemination and embryo transfer increase breeding success in captive mammals.

[3 marks]

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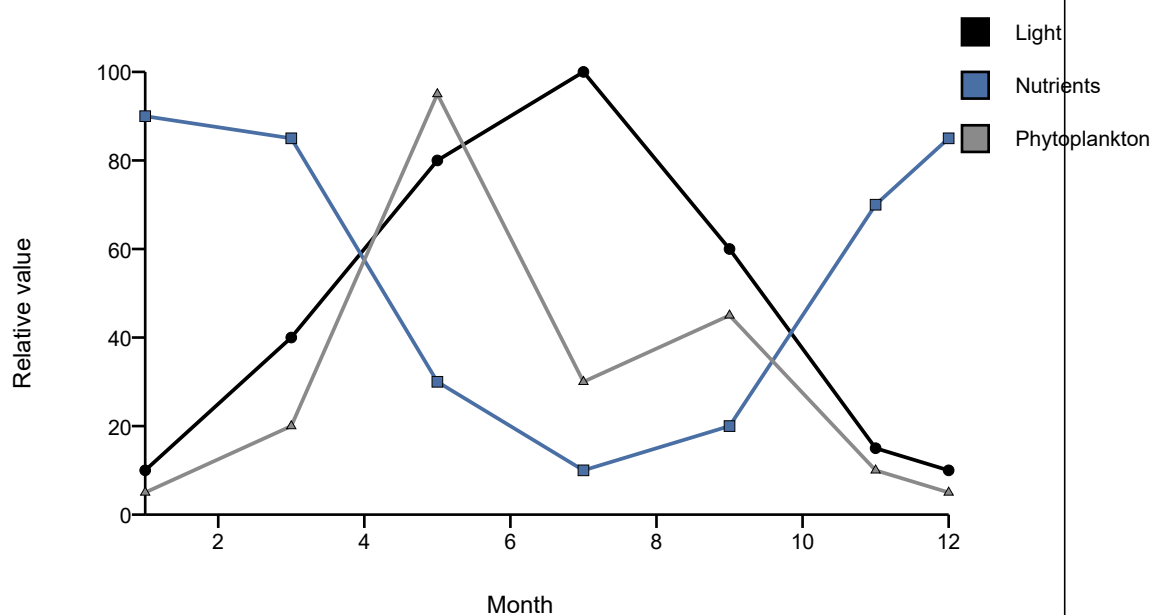
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10

0 5

Figure 1 shows the seasonal changes in light intensity, nutrient concentration and phytoplankton biomass in the surface water of a temperate ocean in the northern hemisphere.

Figure 1



0 5

. 1

Use Figure 1 to explain the spring peak and summer decline in phytoplankton biomass.

[4 marks]

0 5 . 2

Explain why tropical open oceans have low productivity all year despite high light and temperature.

[2 marks]**0 5 . 3**

Explain why areas of upwelling and coastal waters are the most productive parts of the sea.

[3 marks]**0 5 . 4**

Explain why phosphate is often the limiting nutrient in the sea.

[1 mark]**10****0 6**

Table 1 shows typical energy values for an agroecosystem. Values are units of energy remaining from 100 000 units of solar energy.

Table 1

Stage	Energy / arbitrary units
Solar energy reaching the crop	100 000
Energy captured during photosynthesis	4 000
Energy stored in crop biomass	1 000

0 6 . 1

Calculate the percentage of the solar energy reaching the crop that is stored in crop biomass.

[1 mark]

Answer

%

06 . 2

Explain why so little of the solar energy is stored in the crop.

[3 marks]

06 . 3

Explain why few food chains have more than four trophic levels.

[2 marks]

06 . 4

Explain why rearing sheep or cattle is the best option for food production in upland and semi-arid areas.

[2 marks]

06 . 5

Explain how pigs can increase the efficiency of a food production system.

[2 marks]

10

07 . 1

Explain how each of the following changes in abiotic factors caused by human activities can make a habitat unsuitable for species living there.

[6 marks]

Water availability

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Dissolved oxygen

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pH

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07 . 2

Explain how increased water turbidity caused by ploughing, mining or dredging harms aquatic organisms.

[2 marks]

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07 . 3

Explain why species that were already rare are most likely to be lost when abiotic conditions change.

[2 marks]

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10

08 . 1

Explain what a carbon footprint should include when calculated for a car.

[3 marks]

08 . 2

Describe what the WWF Living Planet Index measures and how it is used.

[2 marks]

08 . 3

Describe how the Ol Pejeta reserve in Kenya combines commercial cattle ranching with wildlife conservation.

[5 marks]

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10

09 . 1

Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

- A Close planting of a single non-indigenous species ☐
- B Mixed indigenous species with a mixed age structure ☐
- C Clear-felling on a 40-year rotation ☐
- D Removal of dead wood ☐

09 . 2

Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

1

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2

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3

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4

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5

1 0 . 1

Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

[9 marks]

Do not write outside the box

Extra space

10 . 2

State **one** reason why rainforest conservation is important to countries far from the tropics.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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Do not write outside the box

END OF QUESTIONS